

AWR2E44PBGAMXRQ1

Quality, reliability & packaging data download

Status: ACTIVE

Report date: 06/02/2026



Assembly site: Ext-Mfg

RoHS	Yes
REACH	Yes
Device marking	AWR2E44P, 539A, BG
Lead finish/Ball material	—
MSL rating/Peak reflow	Level-3-260C-168 HR
Rating	Automotive

Material content

Homogeneous Material Level Component Level							
Component	Substance	CAS Number	Amount (mg)	Percentage %	ppm	Percentage %	ppm
Mold Compound							
Other Inorganic Materials	Fused Silica	60676-86-0	137.688901	85.000000	850000	30.831983	308320
Other Nonferrous Metals and Alloys	Metal Hydroxide	Trade Secret	7.289412	4.500000	45000	1.632281	16323
Other Organic Materials	Carbon Black	1333-86-4	0.809935	0.500000	5000	0.181365	1814
Other Organic Materials	Proprietary Non Halide Flame Retardant	Trade Secret	8.099347	5.000000	50000	1.813646	18136
Thermoplastics	Epoxy	85954-11-6	8.099347	5.000000	50000	1.813646	18136
Sub-total	—	—	161.986942	100	1000000	36.272921	362729
Semiconductor Device							
Ceramics / Glass	Doped Silicon	7440-21-3	29.981571	100.000000	1000000	6.713622	67136
Sub-total	—	—	29.981571	100	1000000	6.713622	67136
Solder Ball							
Copper and Its Alloys	Copper	7440-50-8	0.350259	0.500000	5000	0.078432	784
Nickel and Its Alloys	Nickel	7440-02-0	0.035026	0.050000	500	0.007843	78
Other Nonferrous Metals and Alloys	Bismuth	7440-69-9	2.101553	3.000000	30000	0.470590	4706
Other Nonferrous Metals and Alloys	Germanium	7440-56-4	0.004904	0.007001	70	0.001098	11
Other Nonferrous Metals and Alloys	Tin	7440-31-5	64.757957	92.442999	924430	14.500924	145009
Precious Metals	Silver	7440-22-4	2.802071	4.000000	40000	0.627454	6275
Sub-total	—	—	70.051770	100	1000000	15.686341	156863
Solder Bump							
Other Nonferrous Metals and Alloys	Tin	7440-31-5	1.431625	98.199973	982000	0.320577	3206
Precious Metals	Silver	7440-22-4	0.026242	1.800027	18000	0.005876	59
Sub-total	—	—	1.457867	100	1000000	0.326453	3265
Substrate							
Ceramics / Glass	Random E-Glass	65997-17-3	58.433619	31.913500	319135	13.084747	130847
Copper and Its Alloys	Copper	7440-50-8	58.633564	32.022700	320227	13.129519	131295
Not Categorized	Proprietary Filler	—	8.039189	4.390600	43906	1.800175	18002
Other Inorganic Materials	Aluminum Hydroxide	21645-51-2	0.069944	0.038200	382	0.015662	157
Other Inorganic Materials	Silica	7631-86-9	31.276593	17.081700	170817	7.003610	70036
Other Nonferrous Metals and Alloys	Barium Sulfate	7727-43-7	1.629773	0.890100	8901	0.364947	3649
Thermoplastics	Epoxy	85954-11-6	25.017319	13.663200	136632	5.602002	56020
Sub-total	—	—	183.100001	100	1000000	41.000663	410007
Total	—	—	446.578151	—	—	100	1000000

MTBF/FIT estimates

MTBF / FIT		MTBF / FIT supporting data							
MTBF	FIT	Usage temp (°C)	Conf level (%)	Activation energy (eV)	Test temp (°C)	Test duration (hours)	Sample size	Fails	Additional comments
5.10204082 × 10 ⁸	1.96	55	60	0.7	130	1000	4676	0	—

Qualification summary

Type	AEC Q100 test #	Test spec	Min lot qty	SS / lot	Test name	Condition	Result	Notes
Test group A - accelerated environment stress test								
THB/HAST	A2	JESD22-A101/JESD22-A110	3	77	Biased HAST	130C/85%RH 96 hours	Pass	Or equivalent Q100 condition
AC/UHAST	A3	JESD22-A102/JESD22-A118	3	77	Unbiased HAST	130C/85%RH for 96 hours	Pass	Or equivalent Q100 condition
TC	A4	JESD22-A104	3	77	Temperature cycle	Per grade requirements. See data sheet.	Pass	—
TC-WBP	A4	MIL-STD883 method 2011	1	30	Post temp cycle bond pull	Per requirements	Pass	As applicable per die configuration
HTSL	A6	JESD22-A103	1	45	High temp storage bake	Per grade requirements. See data sheet.	Pass	—
Test group B - accelerated lifetime simulation test								
HTOL	B1	JESD22-A108	3	77	High temperature operating life	Per grade requirements. See data sheet.	Pass	—
ELFR	B2	AEC Q100-008	3	800	Early life failure rate	Per grade requirements. See data sheet.	Pass	—
Test group C - package assembly integrity tests								
WBS	C1	AEC Q100-001	1	30	Wire bond shear	Cpk > 1.67	Pass	As applicable per die configuration
WBP	C2	MIL-STD883 method 2011	1	30	Wire bond pull	Cpk > 1.67	Pass	As applicable per die configuration
SD	C3	JEDEC J-STD-002	1	15	Solderability	>95% lead coverage	Pass	—
PD	C4	JESD22-B100 and B108	3	10	Physical dimensions	Cpk > 1.67	Pass	—
SBS	C5	AEC Q100-010	3	5 balls from 10 devices	Solder Ball Shear	Cpk > 1.67	Pass	As applicable per die configuration
Test group D - die fabrication reliability tests								
EM	D1	—	—	—	Electromigration	Per technology requirements	Pass	—
TDDDB	D2	—	—	—	Time dependent dielectric breakdown	Per technology requirements	Pass	—
HCI	D3	—	—	—	Hot carrier injection	Per technology requirements	Pass	—
BTI	D4	—	—	—	Bias temperature instability	Per technology requirements	Pass	—
SM	D5	—	—	—	Stress Migration	Per technology requirements	Pass	—
Test group E - electrical verification								

HBM	E2	AEC Q100-002	1	3	Electrostatic discharge - human body model	Per AEC Q100-002	See data sheet	—
CDM	E3	AEC Q100-011	1	3	Electrostatic discharge - charged device model	Per AEC Q100-011	See data sheet	—
LU	E4	AEC Q100-004	1	3	Latch-up	Per AEC Q100-004	Pass	As applicable per Q100-004
ED	E5	AEC Q100-009	3	30	Electrical distributions	Per AEC Q100-009	Pass	—

Ongoing reliability monitoring

No results found

Additional resources

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[Restricted chemical test report](#)

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